AUTOMATED SOCIAL SCIENCE · CROSSED MODEL ROLES

GPT scientist × Qw

GPT Sol designed the studies. Local Qwen generated the evidence.

Same four scenarios as Manning, Zhu & Horton — but five new factors per study and no requirement to reuse the paper's causal models.

108 / 108	4 × 27	5	40.2 min	Qwen3-8B
interactions measured	pairwise-balanced cells	three-level factors / study	recorded model-cell runtime	local, thinking disabled

☕ Mug bargaining

deal + price

52% deal rate — **$14.36** mean price when measured

- **Buyer budget · $6 → $18**: 22% → 67% — deal rate; price $8.50 → $18
- **Seller floor · $4 → $16**: 78% → 33% — deal rate fell as predicted

Clear: budget, reservation price, outside options, procedure. **Miss:** attachment was not monotonic.

⚖ Bail setting

$22.2k mean bail — **4 / 5** clear predicted patterns

- **Prior convictions · 0 → 4**: $13.9k → $29.4k — strong monotonic increase
- **Flight risk · low → high**: $17.5k → $31.1k — evidence also rose monotonically

Clear: priors, flight risk, evidence, community ties. **Mixed:** cooperation dipped only at the highest level.

💼 Lawyer interview

hire + 0–100 rating

63% hire rate — **72.4** mean rating

- **Clarity · low → high**: 57 → 81 — hire rate 33% → 78%
- **Bar · not passed → admitted**: 33% → 78% — hire recommendation

Placebo held: height had no monotonic effect. **Mixed:** reasoning and experience peaked at middle levels.

🖼 Art auction

price

$171.9 mean final price — **54%** mean allocative efficiency

- **Bid increment · $10 → $50**: $127 → $242 — efficiency 70% → 28%
- **Private → exact public values**: $212 → $127 — no-sale outcomes rose sharply

Fragile: only 41% efficient allocations; 10 no-sales; 3 winners were measured above their own valuation.

What changed when the scientist model changed?

Descriptive comparison — design

PUBLISHED PAPER
GPT-4 scientist + subjects
1,073 full-factorial cells. More theory-anchored structure; the auction cleanly targeted second-price logic.

EARLIER REPLICATION
Qwen scientist + Qwen subjects
108 cells, 106 valid. Simpler three-factor designs; 9 of 12 predicted directions matched its own simulated evidence.

CURRENT RUN
GPT Sol scientist + Qwen subjects
108 orthogonal-array cells, 108 measured. Richer tests and a placebo protocol compliance remained a major weakness.

WHAT QWEN SAW Only its role, the public scenario and treatment instructions, its own private information, and the public transcript — never GPT's predictions, analysis plan, the paper's findings, or another agent's private prompt.

One local-model draw per cell · Not human behavior · Planner was not blinded to prior project context

Design and run: October 14–15, 2026 · Full prompts a